

Computer Devices and Sustainability

**CSCI 1101 Introduction to
Computer Science**

Definition of SUSTAINABILITY

According to Dictionary.com:

- 1. the ability to be sustained, supported, upheld, or confirmed.
- 2. Environmental Science. the quality of **not** being **harmful** to the **environment** or depleting natural resources, and thereby supporting long-term ecological balance: *“The committee is developing sustainability standards for products that use energy.”*

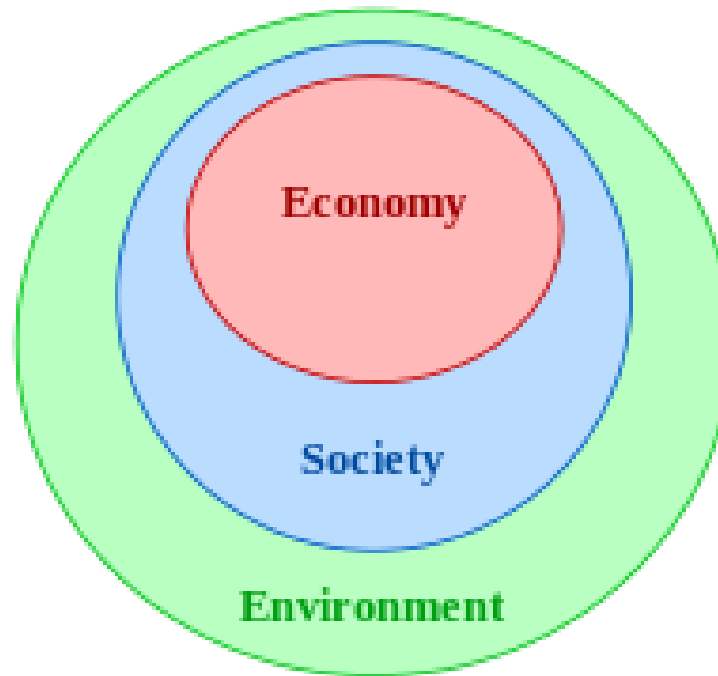
Sustainable Development

Is a process for meeting **human development goals** while **sustaining** the ability of natural systems to continue to provide the **natural resources** and **ecosystem services** upon which the **economy** and **society** depend.

See en.wikipedia.org/wiki/Sustainable_development#Culture

The Three pillars of sustainability

- 1. Economic
- 2. Environmental
- 3. Social



See en.wikipedia.org/wiki/Sustainability

The Economic pillar

The people in the western world are heavy consumers. In fact, we consume far more than our *fair share*. Meanwhile, the people in developing countries are exploding in population and some are aspiring to have high-consumption lifestyles too.

We need a sustainable economic model that ensures fair distribution and efficient allocation of our resources.

This pillar ensures that our economic growth maintains a healthy balance with our ecosystem.

The Environmental pillar

We take our natural resources for granted and sometimes we forget that those resources are *not unlimited*. More importantly, our planet must be protected from corporate exploitation and neglect.

This pillar supports initiatives like:

- renewable energy
- reducing fossil fuel consumption and emissions
- sustainable agriculture and fishing
- organic farming
- tree planting and reducing deforestation
- recycling
- better waste management

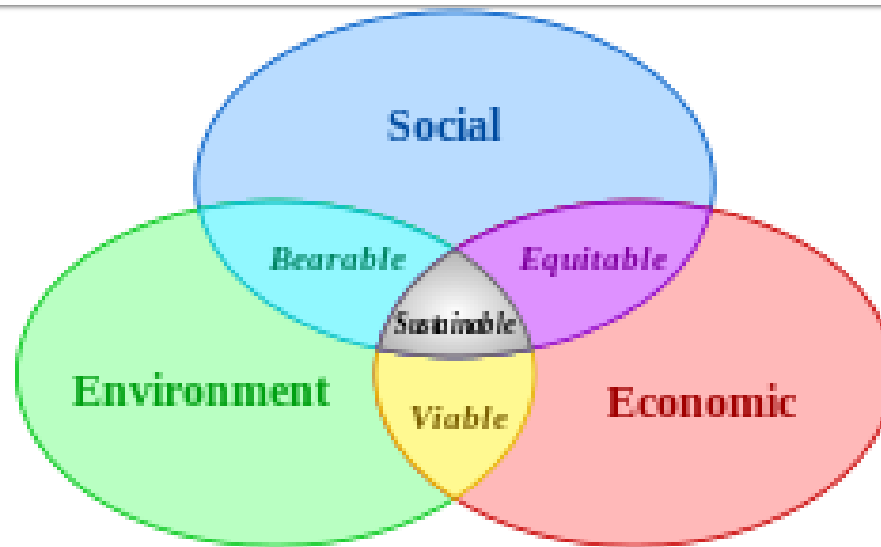
The Social pillar

yadaDROP is a global citizen and you are too. As a global citizen, we must never turn a blind eye to social disruptions that threaten the well-being of people and our environment. We have an ethical responsibility to do something about human inequality, social injustice, and poverty.

This pillar supports initiatives like:

- Peace
- Social justice
- Reducing poverty
- Other grassroots movements that promote social equity

All three pillars are necessary



Viable: a company manufactures recyclable bags that replace plastic bags to protect the environment (good). To be able to make profits, the company uses child labor or some other type of human exploitation (not good at all!).

The fourth pillar

Framing of sustainable development progress according to the **Circles of Sustainability**, used by the **United Nations**.

ECONOMICS

Production & Resourcing
Exchange & Transfer
Accounting & Regulation
Consumption & Use
Labour & Welfare
Technology & Infrastructure
Wealth & Distribution

ECOLOGY

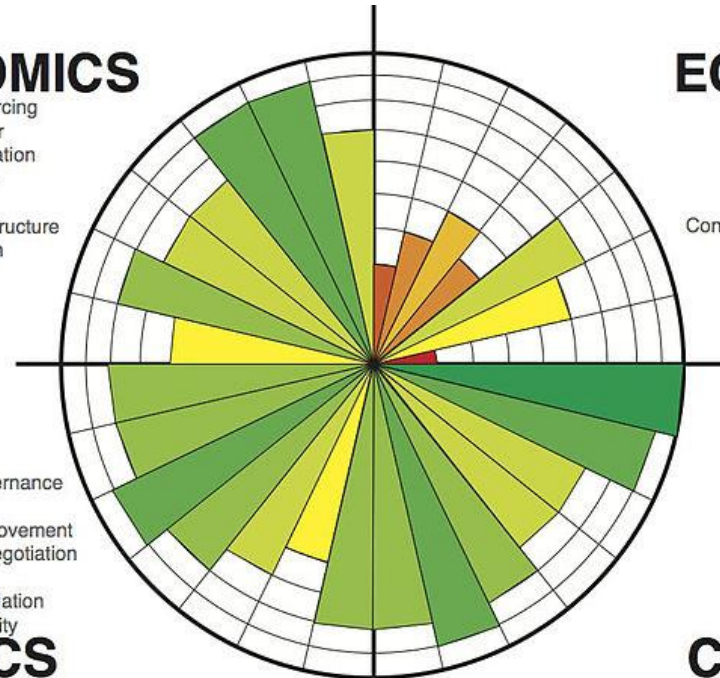
Materials & Energy
Water & Air
Flora & Fauna
Habitat & Food
Place & Space
Constructions & Settlements
Emission & Waste

Organization & Governance
Law & Justice
Communication & Movement
Representation & Negotiation
Security & Accord
Dialogue & Reconciliation
Ethics & Accountability

POLITICS

Engagement & Identity
Recreation & Creativity
Memory & Projection
Belief & Meaning
Gender & Generations
Enquiry & Learning
Health & Wellbeing

CULTURE



CIRCLES OF SUSTAINABILITY

The Culture pillar

- It is needed to reflect the complexity of contemporary society.
- The Circles of Sustainability approach used by Metropolis defines the (fourth) cultural domain as:

Practices, discourses, and material expressions, which, over time, express continuities and discontinuities of social meaning.

The foundational concepts of sustainability

(World Commission on Environment and Development, 1987)

- beauty
- change
- community
- diversity
- ecological health
- equity
- ethics
- interdependence
- limits/scale
- resilience
- systems
- wellbeing

We can use one or more of these foundational concepts as a lens to analyze different situations from the sustainability perspective.

The foundational concepts of sustainability

For example, instead of just accepting that electronic devices produce waste when discarded, we should ask ourselves what are the short- and long-term impacts on people and communities where this garbage is thrown? ([ecological health lens](#)) or is it fair to dump the electronic waste in developing countries that, because of their economic needs, are willing to accept it? ([ethics lens](#))

Computer devices and sustainability

We live in a digital society in which many of our activities are done by means of electronic devices. It is important to realize that the use of electronic devices has certain implications for us and our community.

- Electronic devices use numerous (non-renewable?) resources (silicon, copper, electricity, etc.).
- Electronic devices generate pollutants when discarded (lead, mercury, plastics, etc.).
- Technology can generate inequalities (digital divide).

A few things to reflect about

- What are the advantages and disadvantages of using computer devices?
- How often do you replace your smartphone or computer? Is it really necessary?
- How many devices could we manufacture? [limits/scale]
- How much energy (resources in general) do we spend in manufacturing these devices? [limits/scale]
- How much energy (resources in general) do we spend in using these devices? [limits/scale]
- What type of devices would future generations be able to use? [resilience]
- What happens to old devices? Can this affect our environment? [ecological health] Can this affect other communities? [ethics] [community]
- **What can you do to minimize or eliminate these problems?**
- Can you think of other similar types of issues?